Math 241 Lecture 3

Zipei Nie

University of Illinois Urbana-Champaign

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Outline

1. Review: Dot product and angles
2. Projection of vectors
3. Dot product and planes
4. Angle between planes
5. Lines in 3D
6. Physical application: work
7. Summary

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Dot product: review

Let $\boldsymbol{u} = \langle u_1, u_2, u_3 \rangle$ and $\boldsymbol{v} = \langle v_1, v_2, v_3 \rangle$.

Definition

The *dot product* of $\boldsymbol{u}$ and $\boldsymbol{v}$ is

$$\boldsymbol{u} \cdot \boldsymbol{v} = u_1 v_1 + u_2 v_2 + u_3 v_3.$$

Dot product: review

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Geometric interpretation

$$\boldsymbol{u} \cdot \boldsymbol{v} = |\boldsymbol{u}|\,|\boldsymbol{v}| \cos\theta,$$

where θ is the angle between $\boldsymbol{u}$ and $\boldsymbol{v}$.

Angle between two vectors

Angle formula

For nonzero vectors $\boldsymbol{u}, \boldsymbol{v}$,

$$\cos\theta = \frac{\boldsymbol{u} \cdot \boldsymbol{v}}{|\boldsymbol{u}|\,|\boldsymbol{v}|}.$$

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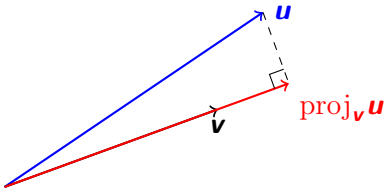
Scalar projection

Scalar projection

The *scalar projection* is the *signed length* of the shadow of $\boldsymbol{u}$ onto the direction of $\boldsymbol{v}$. It is positive if the angle is acute; it is negative if the angle is obtuse.



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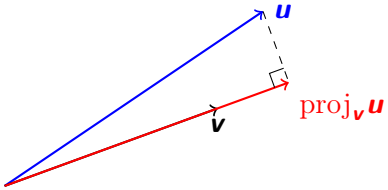

Scalar projection

The *scalar projection* of $\boldsymbol{u}$ onto $\boldsymbol{v}$ is

$$|\boldsymbol{u}|\cos\theta = \cancel{|\boldsymbol{u}|}\frac{\boldsymbol{u}\cdot\boldsymbol{v}}{\cancel{|\boldsymbol{u}|}|\boldsymbol{v}|} = \frac{\boldsymbol{u}\cdot\boldsymbol{v}}{|\boldsymbol{v}|}.$$

Vector projection

unit directional vector

$\frac{\boldsymbol{v}}{|\boldsymbol{v}|}$ has norm one and the same direction with $\boldsymbol{v}$.

Vector projection

The *vector projection* of $\boldsymbol{u}$ onto $\boldsymbol{v}$ is

$$\mathrm{proj}_{\boldsymbol{v}}\,\boldsymbol{u} = \frac{\boldsymbol{u}\cdot\boldsymbol{v}}{|\boldsymbol{v}|}\frac{\boldsymbol{v}}{|\boldsymbol{v}|} = \frac{\boldsymbol{u}\cdot\boldsymbol{v}}{|\boldsymbol{v}|^2}\,\boldsymbol{v} = \frac{\boldsymbol{u}\cdot\boldsymbol{v}}{\boldsymbol{v}\cdot\boldsymbol{v}}\,\boldsymbol{v}.$$

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Example

Project $\boldsymbol{u} = \langle 1, 2, 3\rangle$ onto $\boldsymbol{v} = \langle 1, 1, 0\rangle$.

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Example

Project $\boldsymbol{u} = \langle 1, 2, 3 \rangle$ onto $\boldsymbol{v} = \langle 1, 1, 0 \rangle$.

$$\boldsymbol{u} \cdot \boldsymbol{v} = 3, \quad \boldsymbol{v} \cdot \boldsymbol{v} = 2, \quad \operatorname{proj}_{\boldsymbol{v}} \boldsymbol{u} = \left\langle \frac{3}{2}, \frac{3}{2}, 0 \right\rangle.$$

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Planes via dot product

Question

How to characterize a plane in the 3D space?

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- A point on the plane
- A direction perpendicular to the plane

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Plane equation

Let $\boldsymbol{n} = \langle a, b, c \rangle$ be a vector. The plane through (x_0, y_0, z_0) and perpendicular to $\boldsymbol{n}$ is

$$\boldsymbol{n} \cdot \langle x - x_0,\ y - y_0,\ z - z_0 \rangle = 0.$$

Standard form of a plane

Expanded form

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

Equivalently,

$$ax + by + cz = d.$$

Here $\langle a, b, c \rangle$ is called a normal vector of the plane.

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Example

Find the equation of the plane passing through $(1, 0, 2)$ and perpendicular to $\boldsymbol{n} = \langle 2, -1, 1 \rangle$.

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Example

Find the equation of the plane passing through $(1, 0, 2)$ and perpendicular to $\boldsymbol{n} = \langle 2, -1, 1 \rangle$.

$$2x - y + z = 4$$

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Angle between planes

Key fact

The angle between two planes equals the angle between their normal vectors.

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Formula

If planes have normal vectors $\boldsymbol{n}_1, \boldsymbol{n}_2$, then

$$\cos\theta = \frac{\boldsymbol{n}_1 \cdot \boldsymbol{n}_2}{|\boldsymbol{n}_1|\,|\boldsymbol{n}_2|}.$$

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Lines in 3D: two descriptions

From a point and two normal vectors

A line can be written as the intersection of two non-parallel planes.

Lines in 3D: two descriptions

From a point and two normal vectors

A line can be written as the intersection of two non-parallel planes.

From a point and a direction

A line through point $\boldsymbol{r}_0$ with direction $\boldsymbol{v}$:

$$\boldsymbol{r}(t) = \boldsymbol{r}_0 + t\boldsymbol{v}.$$

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Physical application: work

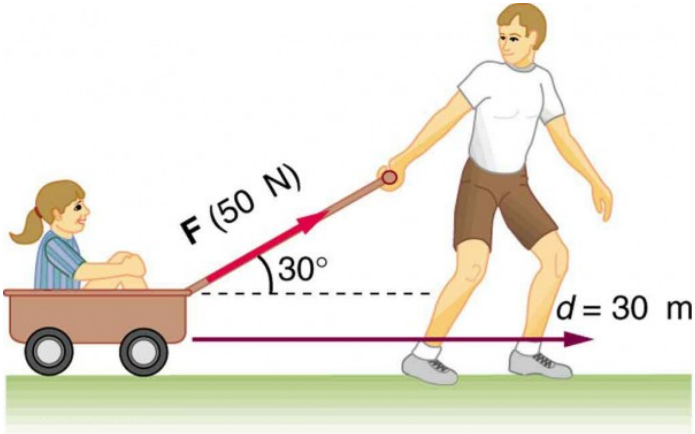

Figure: *

Figure from *College Physics* by Paul Peter Urone and Roger Hinrichs

Work as a dot product

Definition

Work is defined as

$$W = \boldsymbol{F} \cdot \boldsymbol{d},$$

where $\boldsymbol{F}$ is the force and $\boldsymbol{d}$ is the displacement.

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Work is defined as

$$W = \boldsymbol{F} \cdot \boldsymbol{d},$$

where $\boldsymbol{F}$ is the force and $\boldsymbol{d}$ is the displacement.

Remark

Only the component of the force in the direction of motion contributes to the work.

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Takeaways

- Dot product connects algebra and geometry.
- Projections measure components along directions.
- Planes and angles can be described using dot products.
- Lines in 3D admit multiple representations.
- Work in physics is a dot product.